

Linear Algebra – The **Backbone** of **AI/ML**

Linear Algebra – The Backbone

TOPIC :

Problem solving on vector space

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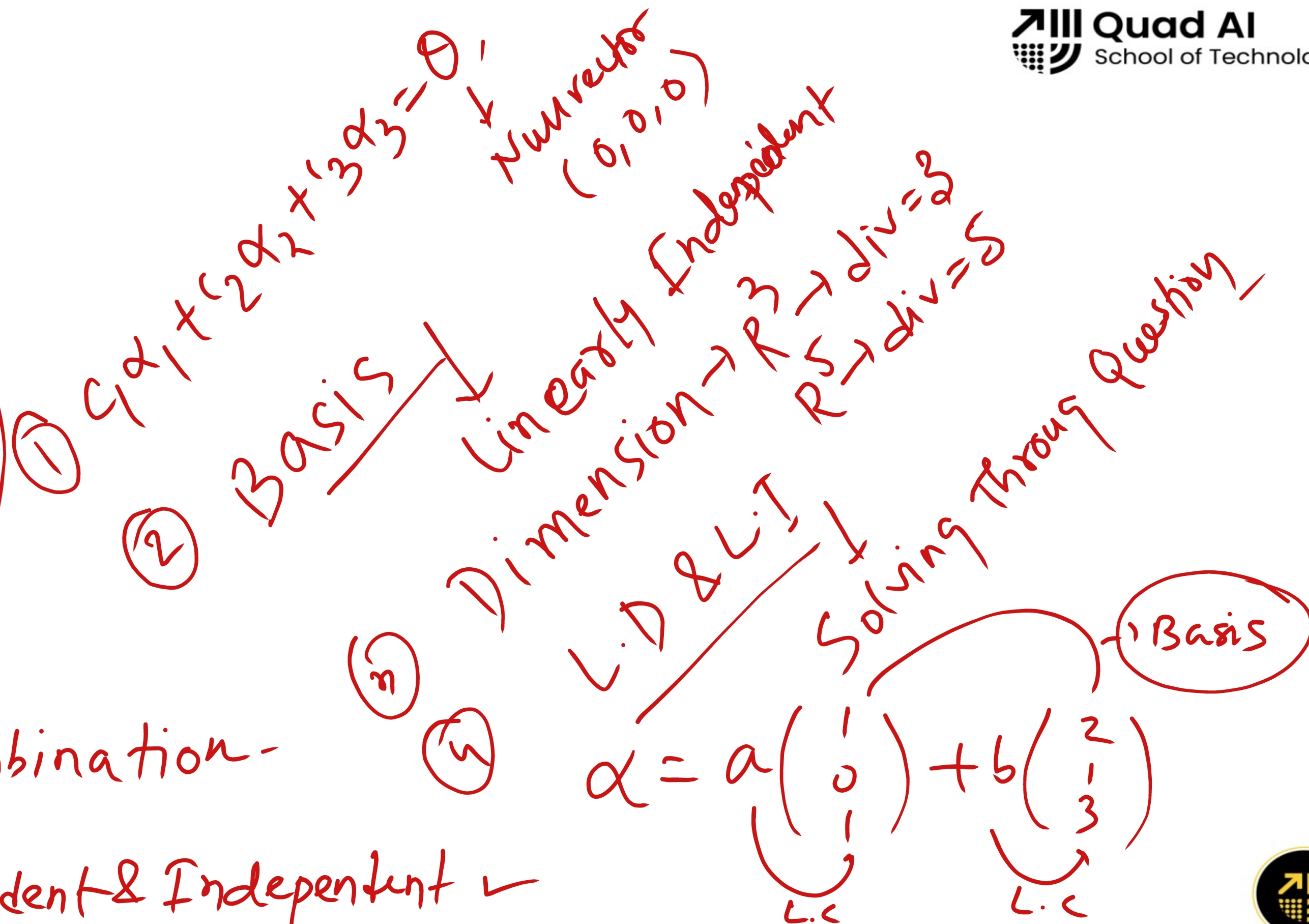
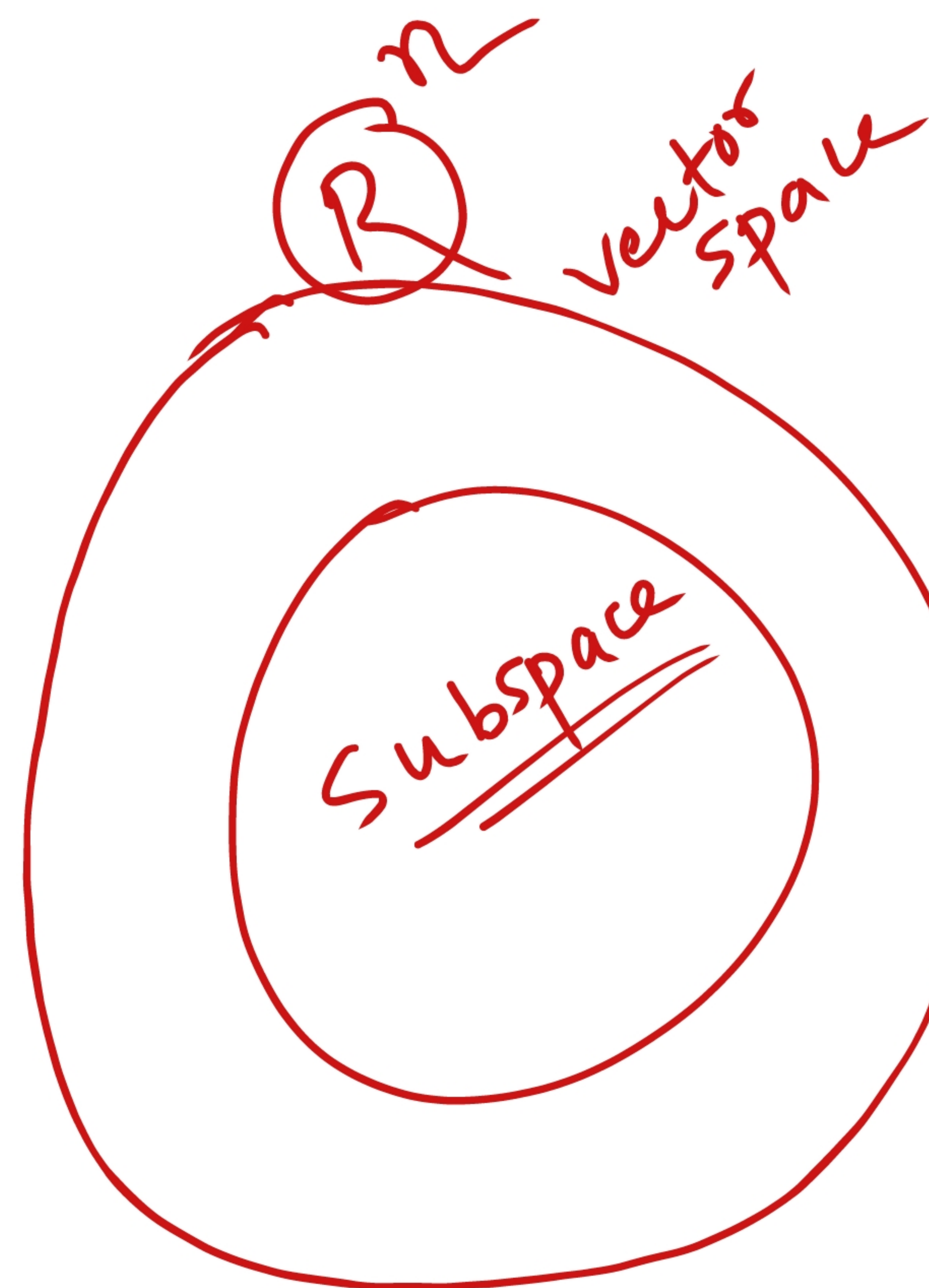
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CONTENT



- ① Problem Solving
Session on vector space
- ② Recap previous lecture
- ③ Cramer's Rule



- ① Linear combination -
- ② Basis
- ③ Linear dependent & Independent ✓

Q. In $\mathbb{R}^3$, $\alpha = (4, 3, 5)$ ✓

value $\left\{ \begin{array}{l} \beta = (0, 1, 3) \\ \gamma = (2, 1, 1) \\ \delta = (4, 2, 2) \end{array} \right.$

Examine-if-

✓ (i) α is a linear combination of β & γ .

✓ (ii) β is a linear combination of γ and δ .

① let $\alpha = c\beta + d\gamma$
Then,
 $(4, 3, 5) = c(0, 1, 3) + d(2, 1, 1)$
 $\Rightarrow (4, 3, 5) = (0+2d, c+d, 3c+d)$
After comparing,
 $2d = 4, \quad c+d = 3, \quad 3c+d = 5$
 $d = 2, \quad c+2 = 3$
giving, $c = 1, \quad d = 2$

Hence, $\alpha = \beta + 2\gamma$

α is a linear combination of β & γ .

P. by Nandini

Check if

$$\beta = \gamma \text{ \& \& } \delta$$

$$\beta = a\gamma + b\delta$$

$$(0, 1, 3) = (2a, a, a) + (4b, 2b, 2b)$$

$$0 = 2a + 4b$$

$$1 = a + 2b$$

$$3 = a + 2b$$

↳ this is not possible

so β is not linear dependent
on γ & δ

Q. Examine if the set of vectors $\{(2,1,1), (1,2,2), (1,1,1)\}$ is linearly dependent in $\mathbb{R}^3$.

Let, $\alpha = (2,1,1)$

$\beta = (1,2,2)$

$\gamma = (1,1,1)$

$c_1 \alpha + c_2 \beta + c_3 \gamma = \mathbf{0} = (0,0,0)$

① At least one non-zero value. $\Rightarrow c_1 (2,1,1) + c_2 (1,2,2) + c_3 (1,1,1) = (0,0,0)$

$$\left. \begin{aligned} 2c_1 + c_2 + c_3 &= 0 \\ c_1 + 2c_2 + c_3 &= 0 \\ c_1 + 2c_2 + c_3 &= 0 \end{aligned} \right\} \begin{aligned} \text{No. of equations} &= 2 \\ \text{No. of variables} &= 3 \end{aligned}$$

$2 < 3$

$$\begin{aligned} 2c_1 + c_2 + c_3 &= 0 \\ c_1 + 2c_2 + c_3 &= 0 \end{aligned}$$

Let $c_3 = k$

$$\begin{aligned} 2c_1 + c_2 &= -k \\ c_1 + 2c_2 &= -k \end{aligned} \quad \times 2$$

$$\begin{aligned} 2c_1 + c_2 &= -k \\ 2c_1 + 4c_2 &= -2k \end{aligned}$$

$$-3c_2 = -k$$

$$c_2 = \frac{k}{3}$$

$$c_1 = -k + \frac{2k}{3}$$

$$c_3 = k$$

$$c_1 = -\frac{k}{3}$$

$$c_2 = \frac{k}{3}, c_3 = k$$

Therefore, given set of vector is linearly dependent in $\mathbb{R}^3$.

② Prove that the set of vectors $\{(1, 2, 2), (2, 1, 2), (2, 2, 1)\}$ is linearly independent in $\mathbb{R}^3$.

Rules:-

$$c_1 = c_2 = c_3 = 0$$

$$c(1, 2, 2) + d(2, 1, 2) + e(2, 2, 1) = (0, 0, 0)$$

$$\therefore c = d = e = 0$$

$\therefore$ It is L.I

$$c + 2d + 2e = 0 \quad \text{--- (i)}$$

$$2c + d + 2e = 0 \quad \text{--- (ii)}$$

$$2c + 2d + e = 0 \quad \text{--- (iii)}$$

$$\text{(i)} - \text{(ii)}, c = d$$

$$\text{(iii)} - \text{(i)}, d = e$$

$$c = d = e$$

put in (i), $d = 0$
 $c = 0$
 $e = 0$

(5) V is the vector space of all 2×2 matrices, prove that the set $S = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \right\}$ is a basis of V . Yes ✓

$$\begin{aligned} \text{Let, } \alpha_1 &= \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \\ \alpha_2 &= \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} \\ \alpha_3 &= \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \\ \alpha_4 &= \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \end{aligned}$$

$$c_1 \alpha_1 + c_2 \alpha_2 + c_3 \alpha_3 + c_4 \alpha_4 = 0$$

$$c_1 \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + c_2 \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} + c_3 \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} + c_4 \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} c_1 + c_2 + c_3 + c_4 & c_2 + c_3 + c_4 \\ c_3 + c_4 & c_4 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\begin{aligned} \therefore c_1 + c_2 + c_3 + c_4 &= 0 \\ c_2 + c_3 + c_4 &= 0 \\ c_3 + c_4 &= 0 \\ c_4 &= 0 \end{aligned}$$

$\therefore c_1 = c_2 = c_3 = c_4 = 0$
Therefore linearly independent.

basis

Friday
Math for AI/ML

offline

1 hour